E18: Exact Lyapunov Descent for the Regularized WCT Curvature Energy

Proof note for Manuscript 2

October 26, 2026

Abstract

We prove a precise conditional version of E18: the regularized Wave Confinement Theory (WCT) curvature-plus-gradient energy is nonnegative and nonincreasing along its exact negative L^2 gradient flow. We compute the full complex variational derivative, including differentiation of the nonlinear curvature weight, and obtain an exact energy–dissipation identity. Periodic boundary conditions and classical solution regularity are imposed to make every integration by parts and chain rule legitimate. The theorem does not establish global existence of the flow or identify it with the independently proposed physical-time WCT dynamics.

1 Definition and assumptions

Let $\Omega = \mathbb{T}^d$ be a flat d-dimensional torus, $1 \le d \le 3$. Fix $c_1, c_2 \ge 0$, $\varepsilon > 0$, and $\alpha \in \mathbb{R}$. For a complex field $\psi : \Omega \to \mathbb{C}$ set

$$\rho = |\psi|^2, \tag{1}$$

$$D(\rho) = \rho + \varepsilon^2 e^{-2\alpha\rho}, \tag{2}$$

$$R_\varepsilon(\psi) = \frac{\overline{\psi}}{D(|\psi|^2)}, \tag{3}$$

$$\Theta_\varepsilon[\psi] = -(\Delta\psi) R_\varepsilon(\psi), \tag{4}$$

$$F(\rho) = \frac{\rho}{D(\rho)^2}. \tag{5}$$

The energy is

$$\boxed{\mathcal{E}[\psi] = \int_\Omega \left(c_1 |\nabla\psi|^2 + c_2 |\Theta_\varepsilon[\psi]|^2 \right) dx = \int_\Omega \left(c_1 |\nabla\psi|^2 + c_2 F(|\psi|^2) |\Delta\psi|^2 \right) dx.} \tag{6}$$

All variational derivatives below use the convention

$$\mathrm{D}\mathcal{E}(\psi)[h] = 2\,\mathrm{Re} \int_\Omega \frac{\delta \mathcal{E}}{\delta \overline{\psi}} \overline{h}\, dx, \tag{7}$$

where h is a complex variation and the underlying tangent space is real. In particular, $\delta\mathcal{E}/\delta\overline{\psi}$ is one-half of the real-L^2 Hilbert gradient for the inner product $(u, v)_{\mathbb{R}} = \mathrm{Re} \int_\Omega u\overline{v}\, dx$.

Proposition 1 (Well-defined, nonnegative energy)**.** *The denominator $D(\rho)$ is strictly positive for every $\rho \ge 0$. For every $\psi \in H^2(\Omega; \mathbb{C})$, the functional in (6) is finite and nonnegative.*

Proof. As $\varepsilon > 0$, one has $D(\rho) = \rho + \varepsilon^2 e^{-2\alpha\rho} > 0$. Thus F is smooth and nonnegative on $[0, \infty)$. For $d \le 3$, $H^2(\Omega)$ embeds continuously in $L^\infty(\Omega)$; therefore $F(|\psi|^2)$ is bounded, while $\nabla\psi, \Delta\psi \in L^2(\Omega)$. Both terms in (6) are finite and nonnegative because $c_1, c_2 \ge 0$. □

2 Exact first variation

The derivative of the curvature weight is

$$D'(\rho) = 1 - 2\alpha\varepsilon^2 e^{-2\alpha\rho}, \tag{8}$$

$$F'(\rho) = \frac{D(\rho) - 2\rho D'(\rho)}{D(\rho)^3} \tag{9}$$

$$= \boxed{\frac{\varepsilon^2 e^{-2\alpha\rho}(1 + 4\alpha\rho) - \rho}{\left(\rho + \varepsilon^2 e^{-2\alpha\rho}\right)^3}}. \tag{10}$$

Proposition 2 (Full complex variational derivative)**.** *For $\psi \in H^4(\Omega;\mathbb{C})$, the L^2 representative of the variational derivative in convention (7) is*

$$\boxed{G[\psi] := \frac{\delta \mathcal{E}}{\delta \overline{\psi}} = -c_1\Delta\psi + c_2\Delta\big(F(|\psi|^2)\Delta\psi\big) + c_2 F'(|\psi|^2)\psi|\Delta\psi|^2.} \tag{11}$$

Proof. Let $h \in C^\infty(\Omega;\mathbb{C})$ and $\psi_s = \psi + sh$ for real s. Direct differentiation gives

$$\frac{d}{ds}|\psi_s|^2\Big|_{s=0} = \psi\overline{h} + \overline{\psi}h, \tag{12}$$

$$\frac{d}{ds}|\Delta\psi_s|^2\Big|_{s=0} = \Delta h\,\Delta\overline{\psi} + \Delta\psi\,\Delta\overline{h}. \tag{13}$$

For the gradient part, periodic integration by parts yields

$$\mathrm{D}\Big(c_1\|\nabla\psi\|_2^2\Big)[h] = 2c_1\,\mathrm{Re}\int_\Omega \nabla\psi\cdot\nabla\overline{h}\,dx = -2c_1\,\mathrm{Re}\int_\Omega \Delta\psi\,\overline{h}\,dx. \tag{14}$$

For the curvature part the product rule gives

$$\begin{aligned}&\mathrm{D}\left(c_2\int_\Omega F(|\psi|^2)|\Delta\psi|^2 dx\right)[h]\\ &\quad= 2c_2\,\mathrm{Re}\int_\Omega F(|\psi|^2)\Delta\psi\,\Delta\overline{h}\,dx + 2c_2\,\mathrm{Re}\int_\Omega F'(|\psi|^2)\psi|\Delta\psi|^2\overline{h}\,dx.\end{aligned} \tag{15}$$

The Laplacian is self-adjoint on $\mathbb{T}^d$, so

$$\int_\Omega F(|\psi|^2)\Delta\psi\,\Delta\overline{h}\,dx = \int_\Omega \Delta\big(F(|\psi|^2)\Delta\psi\big)\overline{h}\,dx. \tag{16}$$

Combining this with (14) and (15),

$$\mathrm{D}\mathcal{E}(\psi)[h] = 2\,\mathrm{Re}\int_\Omega G[\psi]\overline{h}\,dx. \tag{17}$$

This is (11). The H^4 assumption, together with the Sobolev algebra property in $d \le 3$ and the smoothness of F, ensures that every term of $G[\psi]$ lies in L^2. □

3 Exact gradient-flow dissipation theorem

Theorem 1 (E18, conditional exact Lyapunov descent)**.** *Let* $\psi \in C^1([0,T]; H^4(\mathbb{T}^d;\mathbb{C}))$ *solve the specified negative-gradient evolution*

$$\boxed{\partial_t\psi = -G[\psi] = c_1\Delta\psi - c_2\Delta(F(|\psi|^2)\Delta\psi) - c_2F'(|\psi|^2)\psi|\Delta\psi|^2.} \tag{18}$$

Then, for every $t \in [0,T]$,

$$\boxed{\frac{d}{dt}\mathcal{E}[\psi(t)] = -2\|G[\psi(t)]\|^2_{L^2(\Omega)} = -2\|\partial_t\psi(t)\|^2_{L^2(\Omega)} \le 0.} \tag{19}$$

For any $0 \le t_1 \le t_2 \le T$,

$$\boxed{\mathcal{E}[\psi(t_2)] + 2\int_{t_1}^{t_2} \|G[\psi(s)]\|_2^2\, ds = \mathcal{E}[\psi(t_1)].} \tag{20}$$

In particular, $\mathcal{E}[\psi(t_2)] \le \mathcal{E}[\psi(t_1)]$.

Proof. Since the solution is classical in the stated Sobolev class, the real Banach-space chain rule and Proposition 2 give

$$\frac{d}{dt}\mathcal{E}[\psi(t)] = \mathrm{D}\mathcal{E}(\psi(t))[\partial_t\psi(t)] \tag{21}$$

$$= 2\,\mathrm{Re}\int_\Omega G[\psi(t)]\,\overline{\partial_t\psi(t)}\,dx. \tag{22}$$

Inserting $\partial_t\psi = -G[\psi]$ yields

$$\frac{d}{dt}\mathcal{E}[\psi(t)] = -2\,\mathrm{Re}\int_\Omega G[\psi(t)]\overline{G[\psi(t)]}\,dx \tag{23}$$

$$= -2\int_\Omega |G[\psi(t)]|^2 dx \le 0, \tag{24}$$

proving (19). Integrating over $[t_1,t_2]$ proves (20), and dropping the nonnegative dissipation integral gives monotonicity. □

Corollary 1 (Characterization of zero instantaneous dissipation)**.** *Under the hypotheses of Theorem 1, at each* t *one has*

$$\frac{d}{dt}\mathcal{E}[\psi(t)] = 0 \quad\Longleftrightarrow\quad G[\psi(t)] = 0 \text{ in } L^2 \quad\Longleftrightarrow\quad \partial_t\psi(t) = 0. \tag{25}$$

Thus a nonstationary time slice has strictly negative energy derivative.

Proof. The equivalent conditions follow immediately from (19), the nonnegativity of a squared norm, and the evolution equation (18). □

Corollary 2 (Finite accumulated dissipation)**.** *If the hypotheses of Theorem 1 hold on every finite interval* $[0,T]$ *and the solution exists globally for* $t \ge 0$*, then*

$$\int_0^\infty \|G[\psi(t)]\|_2^2 dt \le \frac{1}{2}\mathcal{E}[\psi(0)]. \tag{26}$$

Proof. By Proposition 1 and (20), $2\int_0^T \|G[\psi(t)]\|_2^2 dt \le \mathcal{E}[\psi(0)]$ for every T. Monotone convergence of the nonnegative time integral proves (26). □

4 Boundary terms and conventions

Remark 1 (Nonperiodic domains)**.** *For a sufficiently smooth bounded domain, the first variation has, in addition to the bulk term in* (11)*, the boundary contribution*

$$B[h] = 2\,\mathrm{Re}\int_{\partial\Omega}[(c_1\partial_n\psi - c_2\partial_n(F(|\psi|^2)\Delta\psi))\overline{h} + c_2F(|\psi|^2)\Delta\psi\,\partial_n\overline{h}]\,dS. \tag{27}$$

The bulk L^2-gradient derivation applies when the admissible variations and boundary conditions make $B[h] = 0$, and the evolution is compatible with those boundary conditions. Imposing an unspecified boundary condition is not enough for a fourth-order integration-by-parts argument.

Remark 2 (Factor of two)**.** *With the convention* (7)*, the exact flow $\psi_t = -\delta\mathcal{E}/\delta\overline{\psi}$ yields the coefficient* 2 *in* (19)*. With the real Hilbert gradient $\nabla_{\mathbb{R}}\mathcal{E} = 2G$, the flow $\psi_t = -\nabla_{\mathbb{R}}\mathcal{E}$ instead obeys $d\mathcal{E}/dt = -\|\nabla_{\mathbb{R}}\mathcal{E}\|_2^2$. The factor is a normalization choice, not a substantive discrepancy.*

5 Scope and limitations

Theorem 1 establishes an *exact conditional* Lyapunov identity for solutions of *the specified gradient flow*. It does not assert: (i) local or global existence for arbitrary initial data; (ii) uniform parabolicity, since $F(0) = 0$; (iii) convergence to a unique equilibrium; (iv) whole-space spatial localization; or (v) derivation of (18) from a distinct Hamiltonian, wave, or mixed dissipative WCT dynamical equation.

For an arbitrary specified evolution $\psi_t = X[\psi]$, the same chain rule gives only

$$\frac{d}{dt}\mathcal{E}[\psi(t)] = 2\,\mathrm{Re}\int_\Omega G[\psi]\,\overline{X[\psi]}\,dx, \tag{28}$$

and this expression does not have a fixed sign without additional compatibility assumptions.
Conclusion. The mathematically precise E18 statement is

$$\boxed{\psi_t = -\frac{\delta\mathcal{E}}{\delta\overline{\psi}} \quad\Longrightarrow\quad \frac{d\mathcal{E}}{dt} = -2\left\|\frac{\delta\mathcal{E}}{\delta\overline{\psi}}\right\|_2^2 \le 0,}$$

for the corrected regularized energy, on the stated class of solutions.

References

[1] R. J. Reyes, *WCT Full Equation List Corrected*, E17–E18, online repository, https://github.com/rickyjreyes/geometry_of_resonance/blob/main/WCT_FULL_EQUATION_LIST_CORRECTED.md.

[2] R. J. Reyes, *WCT Lean Derived Audit*, E18 conditional algebraic descent core, https://github.com/rickyjreyes/wct-lean/blob/main/WCTLean/DerivedAudit.lean.